

OPAMP CIRCUITS

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Abstract

This report includes the details of design and mathematical derivation of an Integrator, Differentiator, and Half wave precision rectifier using op-amp.

1 Integrator

An Op-Amp Integrator is an operational amplifier circuit configuration where the output voltage is proportional to the time-integral of the input voltage. Integrator uses a feed back capacitor (C_f) to store charge over time. In reality the feedback capacitor requires a feedback resistor (R_f) parallel to C_f

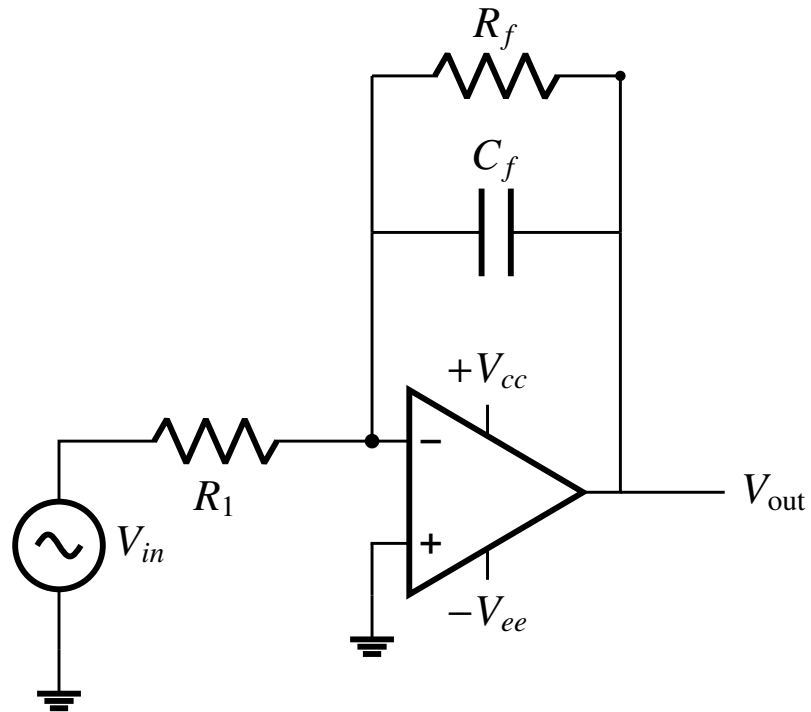
Why we connect a Feedback Resistor (R_f) :

In a theoretical integrator, the feedback loop consists solely of a capacitor C_f . However, in practice, a resistor R_f must be connected in parallel with C_f for the following reasons:

- **DC Stability:** Op-amps are not perfect; they have small internal input bias currents and offset voltages. Without (R_f), the capacitor treats these tiny DC errors as valid signals and integrates them over time.
- **Saturation Prevention:** Because the integral of a DC constant is a ramp ($V = \int k dt$), the output will slowly but surely drift until it hits the positive or negative power supply rail (+15V or -15V). Once it hits the rail, the op-amp saturates and stops reacting to your input signal.
- **Low-Frequency Stability:** It defines a cut-off frequency ($f_c = 1/2\pi R_f C_f$), below which the circuit acts as an amplifier and above which it acts as an integrator.

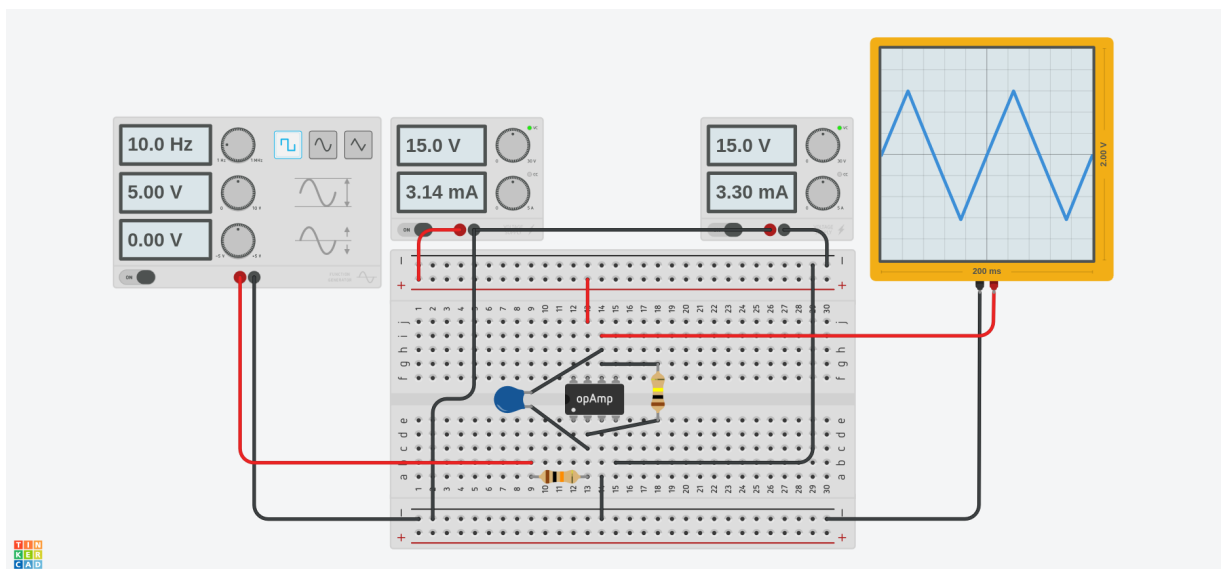
To avoid above problems we need to use R_f parallel to C_f

1.1 Circuit Diagram



Integrator circuit diagram

Simulated circuit diagram



Circuit with square wave as input

1.2 Mathematical Derivation

To derive the relationship between $V_{in}(t)$ and $V_{out}(t)$, we apply Kirchhoff's Current Law (KCL) at the inverting input node.

Node Equations :

Assuming an ideal operational amplifier with infinite open-loop gain, the inverting terminal is at a **virtual ground** (0V). The total current entering the node from the input must equal the total current leaving through the feedback path:

$$I_{in}(t) = I_{feedback}(t) \quad (1)$$

The input current is defined by Ohm's Law:

$$I_{in}(t) = \frac{V_{in}(t) - 0}{R_1} = \frac{V_{in}(t)}{R_1} \quad (2)$$

The feedback current splits into two branches: the current through the resistor (I_{R_f}) and the current through the capacitor (I_{C_f}):

$$I_{feedback}(t) = I_{R_f}(t) + I_{C_f}(t) \quad (3)$$

Using the constitutive equations for a resistor and a capacitor:

$$I_{R_f}(t) = \frac{0 - V_{out}(t)}{R_f} = -\frac{V_{out}(t)}{R_f} \quad (4)$$

$$I_{C_f}(t) = C_f \frac{d(0 - V_{out}(t))}{dt} = -C_f \frac{dV_{out}(t)}{dt} \quad (5)$$

Substituting these into the KCL equation:

$$\frac{V_{in}(t)}{R_1} = -\left[\frac{V_{out}(t)}{R_f} + C_f \frac{dV_{out}(t)}{dt} \right] \quad (6)$$

$$\Rightarrow \frac{dV_{out}(t)}{dt} + \frac{V_{out}(t)}{R_f C_f} = -\frac{V_{in}(t)}{R_1 C_f} \quad (7)$$

Integration Condition :

In a practical integrator, R_f is chosen to be very large. If the rate of change of the signal is fast enough (i.e., the integration time is short compared to the time constant $R_f C_f$), the term $\frac{V_{out}(t)}{R_f C_f}$ becomes negligible (≈ 0). The equation (7) simplifies to:

$$\frac{dV_{out}(t)}{dt} \approx -\frac{V_{in}(t)}{R_1 C_f} \quad (8)$$

By integrating both sides with respect to time from 0 to t :

$$\boxed{V_{out}(t) = -\frac{1}{R_1 C_f} \int_0^t V_{in}(\tau) d\tau + V_{out}(0)} \quad (9)$$

Note :

- $V_{out}(0)$ represents the initial condition on the capacitor voltage i.e if capacitor is initially uncharged then $V_{out}(0) = 0V$ otherwise $V_{out}(0) = \frac{Q_0}{C_f}$ where Q_0 is the initial charge on the capacitor

1.3 Frequency limit

An integrator behaves well only within a specific frequency band. To obtain a clear waveform the operating frequency f must be constrained by the lower cutoff frequency of the feedback network and the upper bandwidth limits of the operational amplifier.

- The frequency f_a at which the circuit stops integrating and starts acting as a constant-gain inverting amplifier is:

$$f_a = \frac{1}{2\pi R_f C_f} \quad (10)$$

- For clear output, the operating frequency f must be significantly higher than f_a . Generally :

$$f \geq 10 \times f_a \quad (11)$$

1.4 Input vs Output Waveforms

- Square Wave Input: Results in a Triangular Wave output .
- Sine Wave Input: Results in a Cosine Wave output.
- Triangular Wave Input: Results in a Parabolic Wave output.

1.5 Experimental Outputs

- The values of resistors and capacitors we used are $R_1 = 10K\Omega$, $R_f = 100K\Omega$, and $C_f = 10\mu F$

- From the above values

$$R_f C_f = 1 \quad (12)$$

- The desired frequency range is given by

$$f_a = \frac{1}{2\pi R_f C_f} \quad (13)$$

$$\Rightarrow f_a = \frac{1}{2\pi} \quad (14)$$

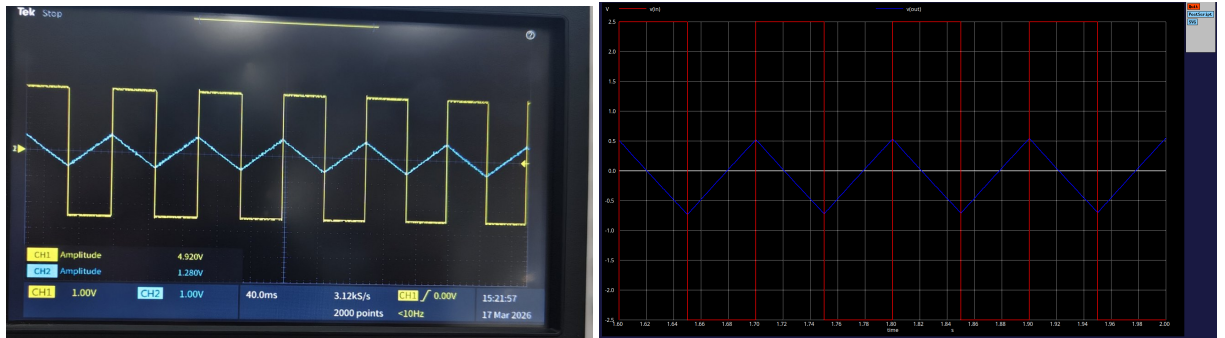
$$\Rightarrow f_a = 0.159Hz \quad (15)$$

$$f > 10f_a \quad (16)$$

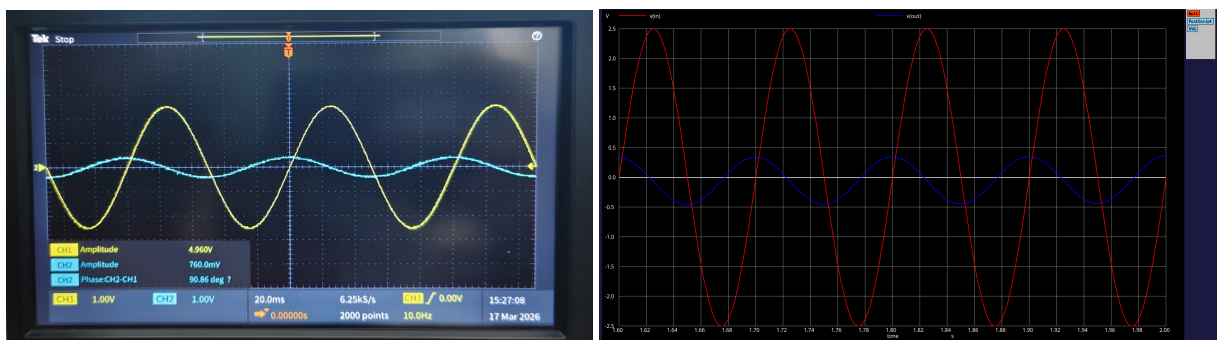
$$\Rightarrow f > 1.59Hz \quad (17)$$

- From the above equation we have taken the frequency of all the different tested input waveforms as 10Hz

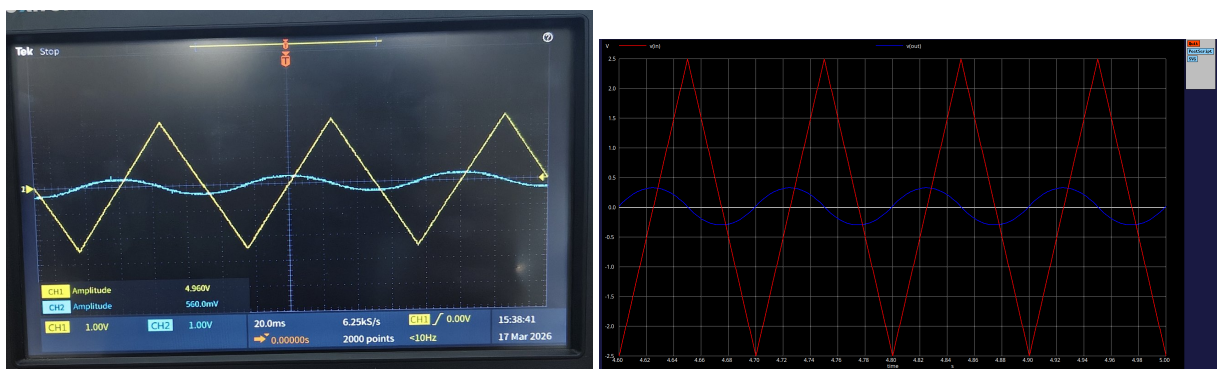
Waveforms



Output waveform with input as square wave



Output waveform with input as sine wave

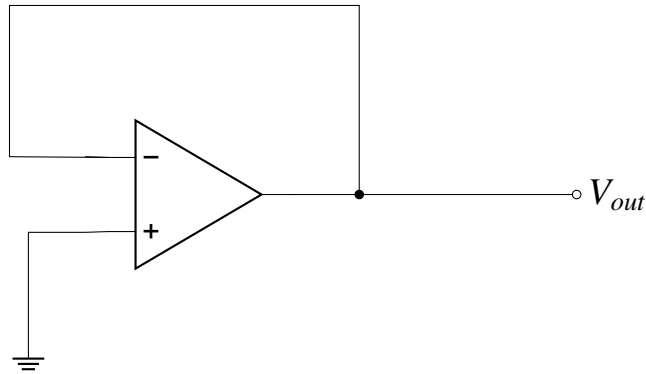


Output waveform with input as triangular wave

1.6 Offset

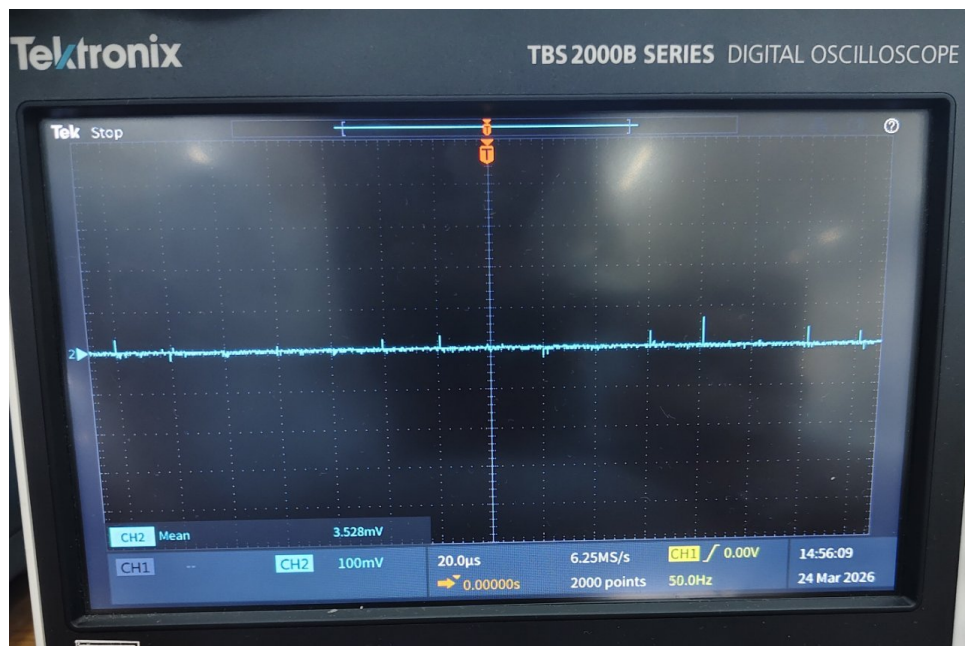
Ideal op-amps have 0 offset i.e when $V_+ = 0$ and V_- is connected to the V_{out} for an ideal case the output should be 0V.

But in practical every op-amp has an offset which can be found in the way :



Circuit diagram to find offset

- Connect the V_+ of the op-amp to the ground and V_- of the op-amp to the output
- Measure the value of V_{out} in this case by enabling mean using the measure menu in the DSO
- The value of V_{out} in this case is the offset of the opamp



Offset measured experimentally

1.7 Theoretical calculations including offset

Real-world op-amps have an input offset voltage. This acts as a constant DC signal at the non-inverting terminal. In a practical integrator, the output offset is governed by the non-inverting gain:

$$V_{out(Off)} = V_{os} \left(1 + \frac{R_f}{R_1} \right) \quad (18)$$

Values Taken :

- $R_1 = 10 \text{ k}\Omega$
- $R_f = 100 \text{ k}\Omega$
- $C_f = 10 \mu\text{F}$
- $V_{os} = 3.52 \text{ mV}$
- Input: Square wave $\pm 2.5 \text{ V}$ at 10 Hz ($T = 0.1 \text{ s}$)

1.7.1 Shift of the centre of output waveform

The waveform will be centered at a DC level determined by the offset:

$$V_{shift} = 3.52 \text{ mV} \left(1 + \frac{100\text{k}}{10\text{k}} \right) = 3.52 \times 11 = \mathbf{38.72 \text{ mV}} \quad (19)$$

1.7.2 General Expression

For the interval $0 \leq t \leq 0.05 \text{ s}$:

$$V_{out}(t) = -\frac{1}{R_1 C_f} \int_0^t V_{in}(\tau) d\tau + V_{out}(0) + V_{shift} \quad (20)$$

Given $V_{out}(0) = 0 \text{ V}$ (initial capacitor state), $V_{in} = 2.5 \text{ V}$, and $V_{shift} = 0.0387 \text{ V}$:

$$V_{out}(t) = -\frac{2.5}{0.1} t + 0 + 0.0387 \quad (21)$$

$$V_{out}(t) = -25t + 0.0387 \quad (22)$$

1.7.3 Output Amplitude

Since $R_f C_f = 1 \text{ s}$ (which is $\gg T/2 = 0.05 \text{ s}$), the circuit behaves as a pure integrator. The peak-to-peak output voltage (V_{p-p}) for a square wave is:

$$V_{p-p} = \frac{V_{in(p-p)}}{R_1 C_f} \times \frac{T}{2} \quad (23)$$

Substituting the values:

$$V_{p-p} = \frac{2.5}{10,000 \times 10 \times 10^{-6}} \times 0.05 = \frac{2.5}{0.1} \times 0.05 = \mathbf{1.25\ V} \quad (24)$$

The amplitude of the output waveform is given by

$$V_{\text{out(amp)}} = 1.25V \quad (25)$$

1.7.4 Conclusion :

The resulting waveform observed on the DSO will be a **Triangular Wave** with the following characteristics:

- Peak-to-Peak observed = 1.28 V

Error :

$$\text{Error} = \frac{1.28 - 1.25}{1.25} \times 100\% \quad (26)$$

$$\Rightarrow \text{Error} = \frac{0.03}{1.25} \times 100 = 2.4\% \quad (27)$$

- **DC Offset (Mean):** $\approx +38.7\ mV$

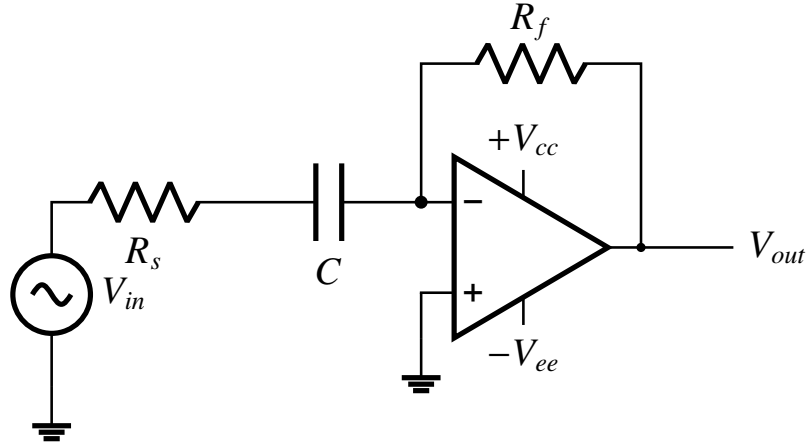
2 Differentiator

An op-amp differentiator is a circuit configuration where the output voltage is directly proportional to the rate of change of the input voltage. Mathematically, it performs the operation of differentiation with respect to time.

Note :

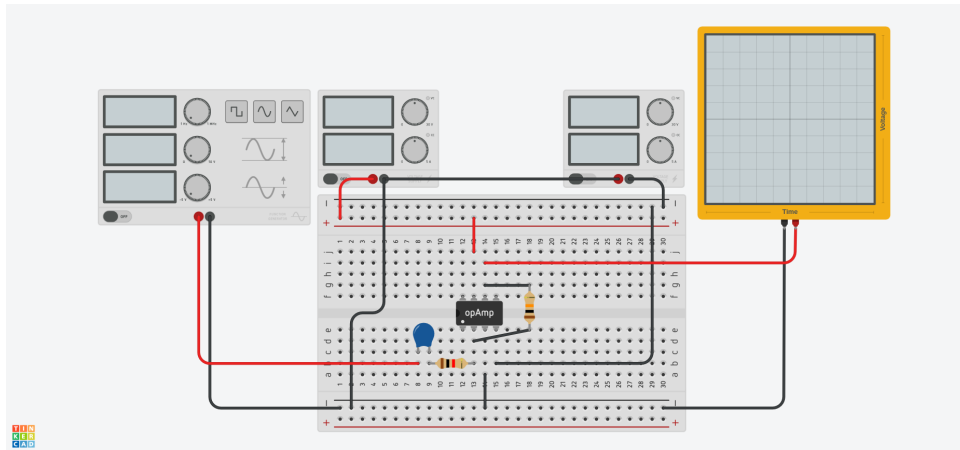
- Ideal differentiator has only capacitor(C) connected to the input and feedback resistor(R_f)
- In general we connect a resistance(R_s) in series with capacitor for the following reasons
- **Stability and Oscillation :** At high frequencies, the reactance of the capacitor ($X_C = 1/2\pi fC$) approaches zero. This causes the closed-loop gain to increase with frequency. This leads to instability and high-frequency oscillations. R_s limits the maximum gain to $A_{max} = R_f/R_s$, stabilizing the loop.
- **Noise :** When we performed the experiment without R_s the output is very badly affected by the noise. To prevent this we added a resistance in series with the capacitance

2.1 Circuit Diagram



Differentiator circuit diagram

Simulated circuit Diagram



Simulated circuit diagram

2.2 Mathematical Derivation of the Practical Differentiator

To derive the relationship between $V_{in}(t)$ and $V_{out}(t)$ for a practical differentiator, we apply Kirchhoff's Current Law (KCL) at the inverting input node.

Node Equations:

Assuming an ideal operational amplifier with infinite open-loop gain, the inverting terminal is at a **virtual ground** (0V). By KCL, the current entering the inverting node through the input branch must equal the current leaving through the feedback resistor:

$$I_{in}(t) = I_f(t) \quad (28)$$

The feedback current through the resistor R_f is defined by Ohm's Law:

$$I_f(t) = \frac{0 - V_{out}(t)}{R_f} = -\frac{V_{out}(t)}{R_f} \quad (29)$$

The input branch consists of a capacitor C and a series resistor R_s . The total voltage drop across this branch is $V_{in}(t) - 0$. The relationship between the current $I_{in}(t)$ and the voltages across these components is:

$$V_{in}(t) = I_{in}(t)R_s + \frac{1}{C} \int I_{in}(t) dt \quad (30)$$

Taking the derivative of Equation (3) with respect to time:

$$\frac{dV_{in}(t)}{dt} = R_s \frac{dI_{in}(t)}{dt} + \frac{I_{in}(t)}{C} \quad (31)$$

Differentiation Condition:

In a practical differentiator, R_s is chosen to be small to ensure stability without significantly altering the desired transfer function. For signal frequencies significantly lower than the corner frequency ($f \ll \frac{1}{2\pi R_s C}$), the voltage drop across the resistor R_s is negligible compared to the drop across the capacitor. Mathematically, the term $R_s \frac{dI_{in}(t)}{dt}$ becomes negligible (≈ 0). Above equation simplifies to:

$$I_{in}(t) \approx C \frac{dV_{in}(t)}{dt} \quad (32)$$

Substituting Equations (29) and (32) into the KCL equation (1):

$$C \frac{dV_{in}(t)}{dt} = -\frac{V_{out}(t)}{R_f} \quad (33)$$

Rearranging for $V_{out}(t)$, we obtain the fundamental differentiator equation:

$$\boxed{V_{out}(t) = -R_f C \frac{dV_{in}(t)}{dt}} \quad (34)$$

2.3 Frequency limit

A differentiator behaves well only within a specific frequency band. To obtain a clear waveform the operating frequency f must obey the following constraints.

- **Corner frequency :** The frequency where the reactance of the capacitor C equals the series resistor R_s .

$$f_c = \frac{1}{2\pi R_s C} \quad (35)$$

- For clear output, the operating frequency f must be significantly lower than f_c . Generally :

$$f \ll f_c (\text{atleast 10 times lower}) \quad (36)$$

2.4 Input vs Output Waveforms

The output waveform is the derivative of the input.

- Triangular Wave Input: Results in a Square Wave output .
- Sine Wave Input: Results in a Cosine Wave output.
- Square Wave Input: Results in a wave which has spikes at rising and falling edges and 0 at other times.

2.5 Experimental Outputs

- The values of resistors and capacitors we used are $R_s = 1K\Omega$, $R_f = 10K\Omega$, and $C_f = 5nF$

- From the above values

$$R_s C_f = 5 \times 10^{-6} s \quad (37)$$

- The desired frequency range is given by

$$f_c = \frac{1}{2\pi R_s C_f} \quad (38)$$

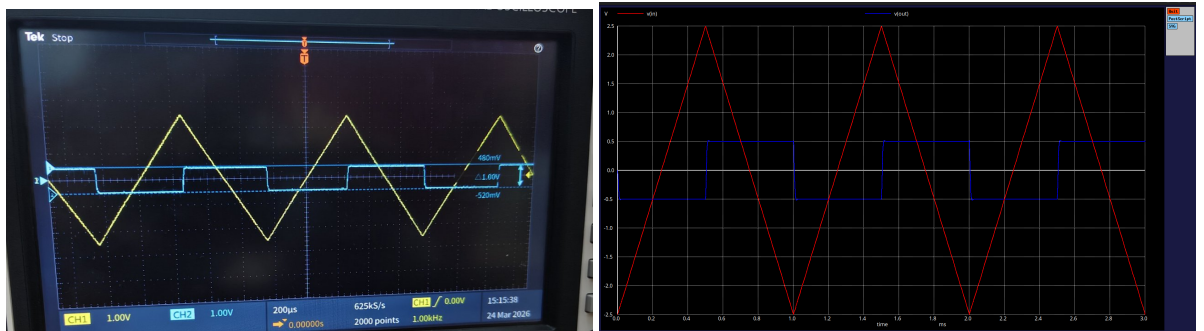
$$\Rightarrow f_c = \frac{1}{\pi} \times 10^5 \quad (39)$$

$$\Rightarrow f_c = 31.82 KHz \quad (40)$$

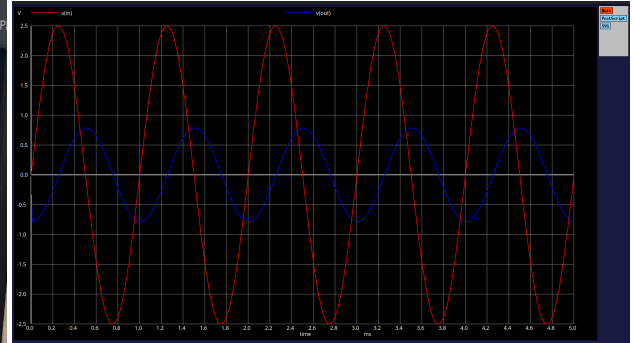
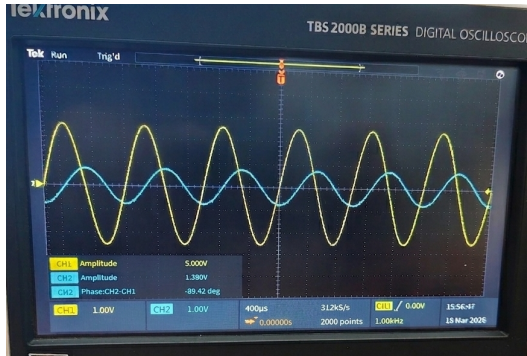
$$\Rightarrow f \ll 31.82 KHz \quad (41)$$

- From the above equation we have taken the frequency of all the different tested input waveforms as 1KHz

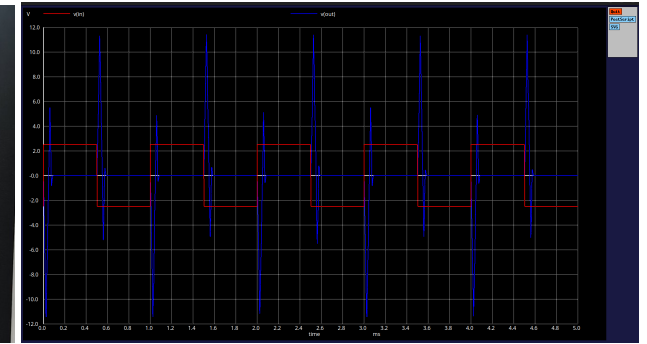
Waveforms



Output waveform with input as triangular wave



Output waveform with input as sine wave



Output waveform with input as square wave

2.6 Theoretical calculations including offset

Values Taken :

- $R_s = 1\text{ k}\Omega$
- $R_f = 10\text{ k}\Omega$
- $C = 5\text{ nF}$
- $V_{os} = 3.52\text{ mV}$ (Measured Op-Amp Offset)
- Input: Triangle wave $\pm 2.5\text{ V}$ ($V_{p-p} = 5\text{ V}$) at 1 kHz ($T = 1\text{ ms}$)

2.6.1 Shift of the centre of output waveform

At DC (0 Hz), the capacitor C is an open circuit. The circuit acts as a voltage follower for the internal offset V_{os} .

$$V_{shift} = V_{os} \times 1 = 3.52\text{ mV} \quad (42)$$

2.6.2 General expression and output amplitude

The output of the practical differentiator is the sum of the differentiated signal and the DC offset shift. Given that the input is a triangle wave, the output is a square wave shifted by the op-amp's internal offset voltage.

Slope Calculation (dV_{in}/dt):

For a 1 kHz triangle wave ($T = 1\text{ ms}$), the voltage ramps from -2.5 V to $+2.5\text{ V}$ over a half-period ($\Delta t = 0.5\text{ ms}$). The rate of change is:

$$\frac{dV_{in}}{dt} = \frac{\Delta V}{\Delta t} = \frac{5\text{ V}}{0.5 \times 10^{-3}\text{ s}} = 10,000\text{ V/s} \quad (43)$$

Output Voltage at Time t :

The general expression for the output, including the measured offset $V_{os} = 3.52\text{ mV}$, is:

$$V_{out}(t) = -R_f C \frac{dV_{in}(t)}{dt} + V_{os} \quad (44)$$

Substituting the circuit parameters ($R_f = 10\text{ k}\Omega$, $C = 5\text{ nF}$) for the rising edge of the input:

$$V_{out}(t) = -(10,000\Omega \times 5 \times 10^{-9}\text{ F}) \times (10,000\text{ V/s}) + 0.00352\text{ V} \quad (45)$$

$$V_{out}(t) = -0.5\text{ V} + 0.00352\text{ V} = \mathbf{-0.49648\text{ V}} \quad (46)$$

During the falling edge of the input ($dV_{in}/dt = -10,000\text{ V/s}$), the output becomes:

$$V_{out}(t) = -(50 \times 10^{-6}) \times (-10,000) + 0.00352\text{ V} = \mathbf{0.50352\text{ V}} \quad (47)$$

Waveform Amplitude and Centering:

The amplitude of the output square wave is defined as the peak deviation from the center. Due to the offset, the positive and negative peaks are slightly asymmetrical relative to 0V:

- $V_{peak(+)} = 503.52\text{ mV}$
- $V_{peak(-)} = -496.48\text{ mV}$

The peak-to-peak amplitude remains:

$$V_{p-p} = V_{peak(+)} - V_{peak(-)} \approx \mathbf{1.0\text{ V}} \quad (48)$$

The peak to peak voltage of the output wave is $\approx 1\text{ V}$, but the entire waveform is vertically translated by the shift $V_{shift} = 3.52\text{ mV}$.

2.6.3 Conclusion

- Theoretical values matches exactly with the output waveform properties in DSO

3 Half Wave Precision Rectifier

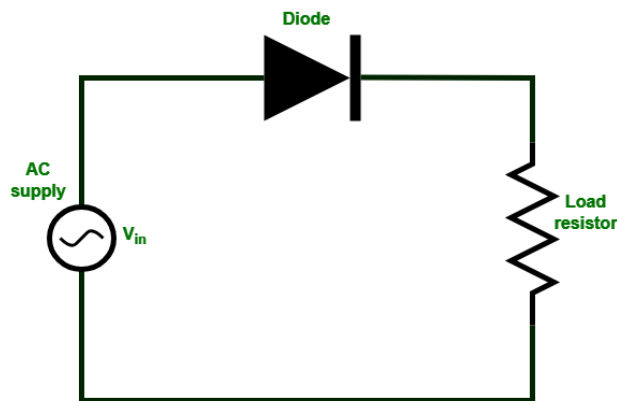
A standard diode-based half-wave rectifier cannot rectify signals smaller than the diode's forward voltage ($V_\gamma \approx 0.7V$). If the input signal is $100mV$, a standard rectifier produces $0V$ output. The Precision Rectifier utilizes the high open-loop gain of an operational amplifier to eliminate this making the diode behave as if it were ideal ($V_\gamma = 0V$).

3.1 Half wave Rectifier

A basic half-wave rectifier consists of an alternating current (AC) source (mainly but not necessary), a single semiconductor diode (D), and a load resistor (R_L). The circuit converts AC voltage into a pulsating DC voltage by allowing current to flow in only one direction.

- **Positive Half-Cycle:** When the input voltage V_{in} is positive and exceeds the diode's barrier potential ($V_\gamma \approx 0.7V$ for Silicon), the diode is **forward-biased**. It conducts current, and the output voltage follows the input minus the diode drop.
- **Negative Half-Cycle:** When V_{in} becomes negative, the diode is **reverse-biased**. It acts as an open circuit (infinite resistance), blocking all current. The output voltage across R_L is $0V$.

3.1.1 Circuit Diagram



Half wave rectifier

3.1.2 Output Equations

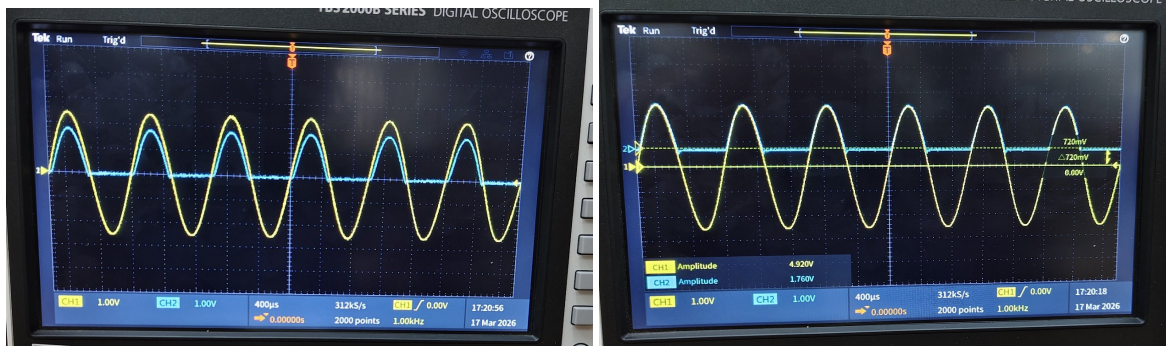
Let the input voltage be defined as:

$$V_{in}(t) = V_m \sin(\omega t) \quad (49)$$

In a real-world circuit, the diode requires a minimum voltage to overcome the depletion region ($V_\gamma \approx 0.7V$). The output only appears when $V_{in} > V_\gamma$:

$$V_{out}(t) = \begin{cases} V_m \sin(\omega t) - V_\gamma & \text{when } V_{in}(t) > V_\gamma \\ 0 & \text{when } V_{in}(t) \leq V_\gamma \end{cases} \quad (50)$$

3.1.3 Output waveforms



Output waveforms with input as sine wave and the threshold voltage

From the above figures it is clear that the threshold voltage is $\approx 0.7V$ and it blocks the input wave whose value is $< 0.7V$

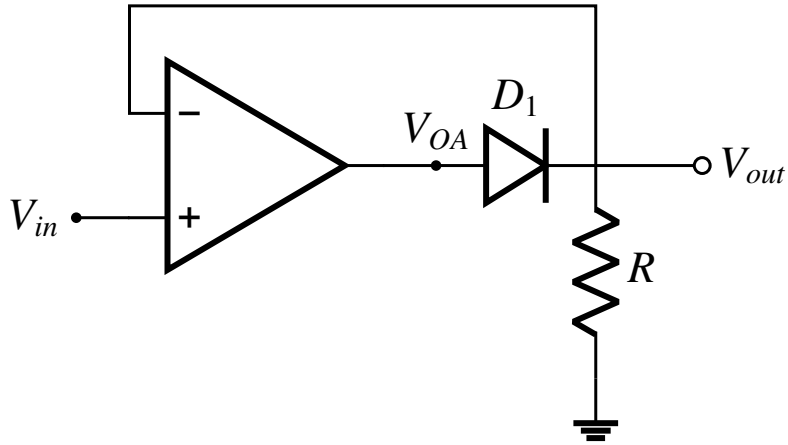
3.2 Difference between Half wave and Half wave precision rectifiers

The basic precision rectifier consists of an op-amp, a diode (D_1) in the feedback loop, and a load resistor (R_L). The input signal is applied to the non-inverting terminal.

The fundamental difference lies in the **placement of the diode relative to the feedback loop**:

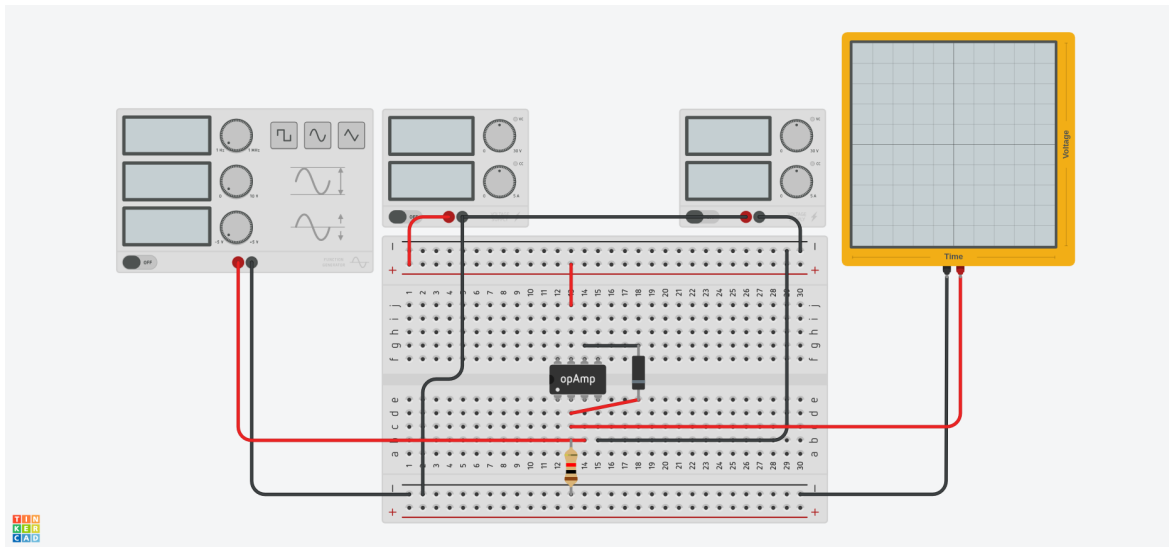
1. **Standard Rectifier:** The diode's $0.7V$ drop is directly subtracted from the input.
2. **Precision Rectifier:** The diode is placed *inside* the feedback loop. The op-amp adjusts its output voltage (V_o) to whatever value is necessary to maintain the virtual short between its input terminals, effectively canceling the V_γ drop.

3.3 Half wave precision rectifier circuit diagram



Half wave rectifier circuit diagram

Simulated Circuit Diagram



Simulated circuit diagram

3.4 Mathematical derivation

Case 1: Positive Half-Cycle ($V_{in} > 0$)

When V_{in} is positive, the op-amp output (V_{OA}) begins to rise. As soon as V_{OA} exceeds V_γ , the diode D_1 becomes forward-biased. The feedback loop closes through the diode to the inverting terminal.

$$V_{out} = V_{OA} - V_\gamma \quad (51)$$

Due to the virtual short ($V_+ = V_-$), the op-amp ensures:

$$V_{out} = V_{in} \quad (52)$$

To achieve this, the internal op-amp output actually sits at $V_{OA} = V_{in} + V_\gamma$. The 0.7V drop exists, but it is inside the loop, so the load only sees V_{in} .

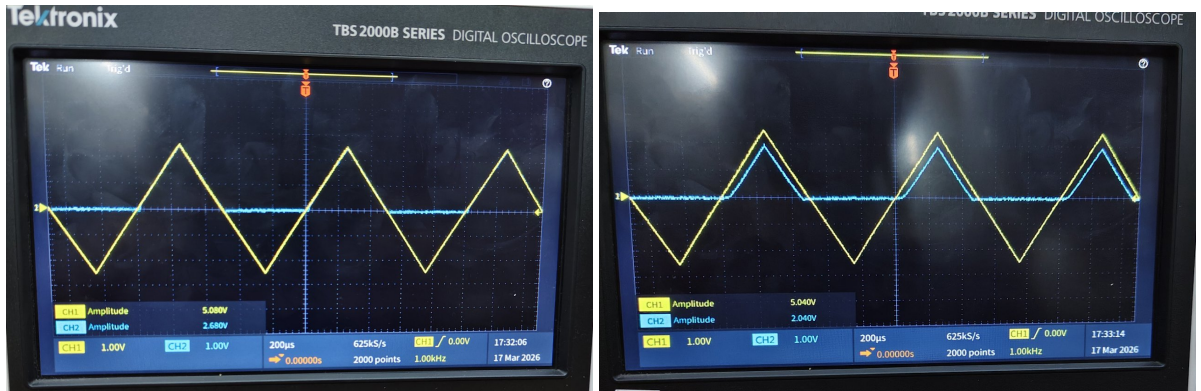
Case 2: Negative Half-Cycle ($V_{in} < 0$)

When V_{in} is negative, the op-amp output V_{OA} swings toward the negative rail. This reverse-biases D_1 , acting as an open circuit.

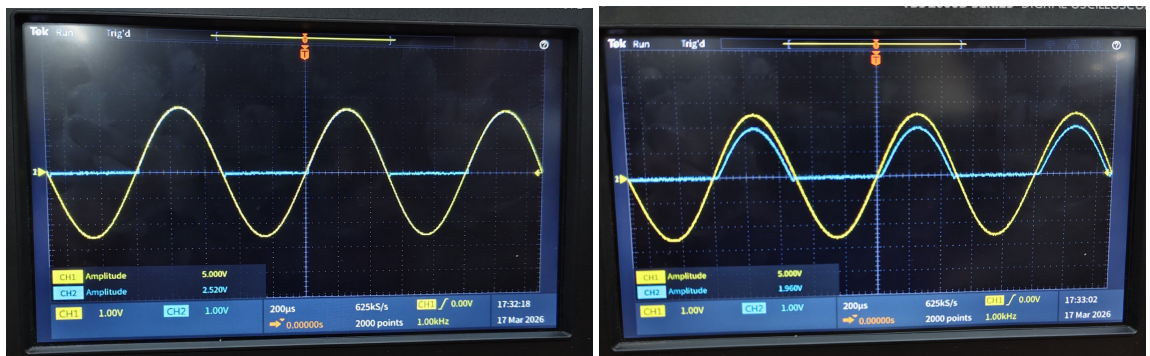
- The feedback loop is broken.
- No current flows through R .
- Therefore:

$$V_{out} = 0V \quad (53)$$

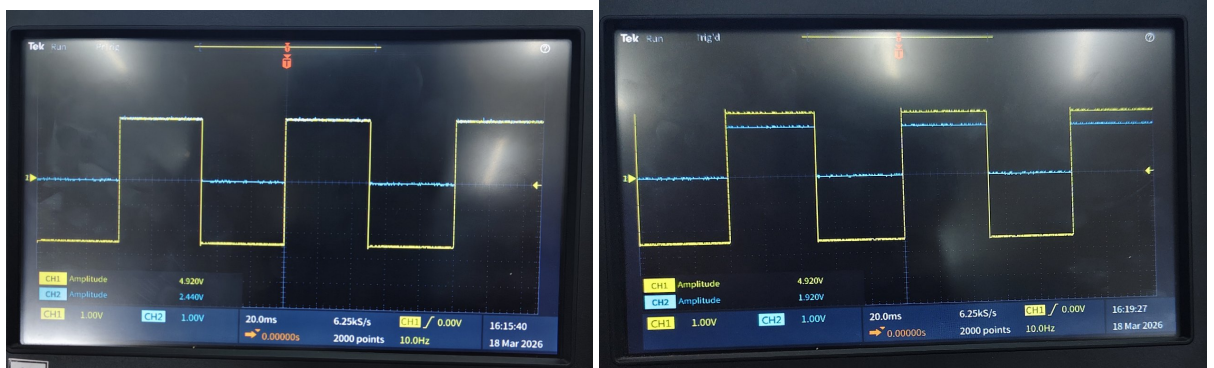
3.5 Output Waveforms



Output waveform with input as triangular wave



Output waveform with input as sine wave



Output waveform with input as square wave

Note :

- In all the above 3 cases the 1st photo is the output when the input is given to a half wave precision rectifier
- The 2nd photo corresponds to the output when the input is applied to a half wave rectifier
- We can see clearly the 0.7V difference when we connect the op-amp and when we don't connect the op-amp

4 Conclusion

In this lab experiment we were successfully able to build integrator, differentiator, half wave precision rectifier using op-amp and successfully tested it with different inputs like square, sine, triangular wave and the results observed experimentally almost matches with the theoretical results.